

Second-Order Linear Homogeneous Equations



The auxiliary equation

- 1 Write the auxiliary (characteristic) equation for $y'' - 5y' + 6y = 0$.
 - 2 Write and simplify the auxiliary equation for $2y'' + 4y' - 6y = 0$.
-

Distinct real roots

- 3 Solve $y'' - 5y' + 6y = 0$.
 - 4 Solve $y'' + 2y' - 3y = 0$, $y(0) = 1$, $y'(0) = 1$.
-

Repeated roots

- 5 Solve $y'' - 6y' + 9y = 0$.
 - 6 Solve $y'' + 4y' + 4y = 0$, $y(0) = 2$, $y'(0) = -1$.
-

Complex roots

- 7 Solve $y'' + 4y = 0$.
 - 8 Solve $y'' - 2y' + 5y = 0$.
-

Application: a damped spring

- 9 A mass–spring–damper system satisfies $y'' + 4y' + 5y = 0$. Find the general solution and classify the damping (over/critically/under-damped).
 - 10 Solve $y'' + 6y' + 9y = 0$, $y(0) = 1$, $y'(0) = 0$, and classify the damping.
-

Further practice: distinct real roots

- 11 Solve $y'' - y' - 6y = 0$.
 - 12 Solve $y'' - 4y = 0$.
-

Further practice: repeated roots

13 Solve $y'' - 8y' + 16y = 0$.

14 Solve $y'' + 10y' + 25y = 0$, $y(0) = 0$, $y'(0) = 3$.

Further practice: complex roots

15 Solve $y'' + 9y = 0$.

16 Solve $y'' + 2y' + 10y = 0$.

Reading the discriminant without fully solving

17 Without fully solving, state whether $y'' + 6y' + 9y = 0$ has distinct real, repeated, or complex roots.

18 Without fully solving, classify the roots of $y'' + y' + y = 0$.

Application: an undamped spring

19 An undamped mass-spring system satisfies $y'' + 16y = 0$. Find the general solution and the angular frequency of oscillation.

20 The same spring is released from rest at displacement 2: $y(0) = 2$, $y'(0) = 0$. Find the particular solution.

One more round: mixed root types

21 Solve $y'' - 9y = 0$, $y(0) = 4$, $y'(0) = 0$.

22 Solve $y'' + 2y' + 2y = 0$.

One final pair

23 Solve $y'' - y = 0$.

24 Solve $y'' + 25y = 0$.

25 Solve $y'' - y' - 2y = 0$, $y(0) = 0$, $y'(0) = 3$.

26 Solve $y'' + 2y' + y = 0$.

A closing pair with initial conditions

27 Solve $y'' - 4y' + 13y = 0$.

28 Solve $y'' + 3y' = 0$, $y(0) = 5$, $y'(0) = -6$.